

WORKED KEY

This practice is built exactly like the real test — same questions, same order, different numbers. Work every problem, then check the worked key. Anything you miss is exactly what to study.

1. Find the sum. Write your answer in standard form: $(5x^3 - 3x^2 + 2x - 7) + (2x^3 + 6x^2 - 9x + 4)$

Answer: $7x^3 + 3x^2 - 7x - 3$

2. Find the difference. Write your answer in standard form: $(8x^2 - 5x + 3) - (3x^2 + 6x - 9)$

Answer: $5x^2 - 11x + 12$

3. Find the product in standard form: $-5x^2(3x^2 - 2x + 6)$

Answer: $-15x^4 + 10x^3 - 30x^2$

4. Find the product in standard form: $(5x - 2)(2x + 7)$

Answer: $10x^2 + 31x - 14$

5. Find the product in standard form: $(x + 5)(x^2 - 4x + 3)$

Answer: $x^3 + x^2 - 17x + 15$

6. Find the product in standard form: $(4x - 3)^2$

Answer: $16x^2 - 24x + 9$

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7. Factor out the GCF: $21x^3 - 14x^2 + 35x$

Answer: GCF is $7x$: $7x(3x^2 - 2x + 5)$

8. Factor: $x^2 + 13x + 42$

Answer: $(x + 6)(x + 7)$

9. Factor: $x^2 - 5x - 24$

Answer: $(x - 8)(x + 3)$

10. Factor: $3x^2 + 11x + 6$

Answer: $(3x + 2)(x + 3)$

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11. Factor: $36x^2 - 121$

Answer: difference of squares: $(6x + 11)(6x - 11)$

12. Factor, if possible. If the trinomial cannot be factored, identify it as "PRIME":
 $x^2 + 6x + 11$

Answer: PRIME — no factor pair of 11 adds to 6 ($1 + 11 = 12$)

13. A rectangular mural on the gym wall has an area of $4x^2 + 28x + 40$ square feet. Factor the expression completely to find binomials that could represent the length and width. (Hint: check the GCF first.)

Answer: $4(x^2 + 7x + 10) = 4(x + 2)(x + 5)$
— possible dimensions $(4x + 8)$ ft by $(x + 5)$ ft (any correct pairing)

Cumulative Review — ACT Style. Circle the best answer.

14. Simplify: $\sqrt{72}$

(A) $2\sqrt{6}$

(B) **$6\sqrt{2}$**

(C) $3\sqrt{6}$

(D) $9\sqrt{2}$

Answer: $\sqrt{(36 \cdot 2)} = 6\sqrt{2}$

15. A sample decays so its mass is $y = 320(1/2)^x$ grams after x hours. What is the mass after 3 hours?

(E) 160 g

(F) 80 g

(G) **40 g**

(H) 20 g

Answer: $320 \cdot 1/8 = 40$

16. Simplify: $(4x^3)^2$

(A) $8x^6$

(B) **$16x^6$**

(C) $16x^5$

(D) $4x^9$

Answer: square the 4 AND multiply exponents